

Project Design Phase

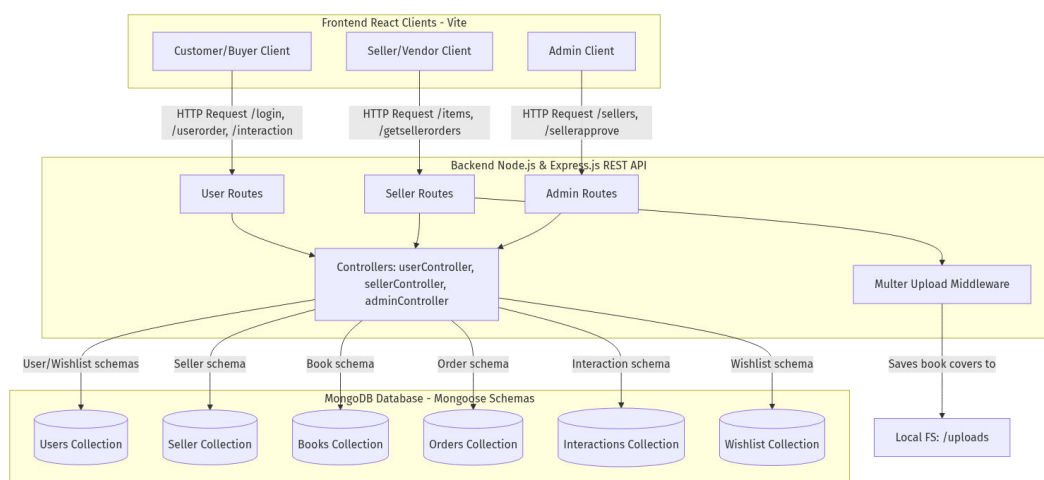
Solution Architecture

Date	June - 25 - 2026
Team ID	SWTID-2026-9984
Project Name	BOOK STORE
Maximum Marks	4 Marks

Solution Architecture: BookEase MERN System

Below is the solution architecture diagram and technical specifications bridging the gap between business requirements and the software system.

Architecture & Data Flow Diagram



Components Summary

1. Presentation Tier: Built using React, Vite, TailwindCSS, and React-Bootstrap. Divided logically into `/User`, `/Seller`, and `/Admin` subfolders. State is managed locally and synced with `localStorage` (for session tokens) and Axios REST requests.

2. Application Tier: Node.js and Express.js backend. Defines controllers that validate client requests and perform transactions. Leverages Multer middleware to process image file uploads.

3. Data Tier: MongoDB instance storing JSON documents mapping to Mongoose schemas. Relationships:

- **User-Book (M:M):** Structured using the `Interaction` collection (storing `readingProgress`, `rating`, and `reviewText`).
- **User-Order (1:M):** Managed by the `Order` collection, referencing `userId`.
- **Book-Seller (1:M):** Books reference their uploading seller (`userId`).